

0. Shapes

#	Layer	Output shape (per image)
1	Conv7x7 s2, 3→32 (+BN+ReLU)	$32 \times S/2 \times S/2$
-	MaxPool 3x3 s2 p1	$32 \times S/4 \times S/4$
2	Conv5x5, 32→64	$64 \times S/4 \times S/4$
3	Conv3x3 s2, 64→128	$128 \times S/8 \times S/8$
4	Conv1x1, 128→256	$256 \times S/8 \times S/8$
5	Conv3x3 s2, 256→256	$256 \times S/16 \times S/16$
6	Conv1x1, 256→512	$512 \times S/16 \times S/16$
head	GAP → Linear 512→256 → ReLU → Linear 256→100	100

1. FLOPs (1 MAC = 2 FLOPs)

General formulas:

- Conv $k \times k$, $C_{in} \rightarrow C_{out}$, output side S_{out} : $FLOPs_{conv} = 2 \cdot [C_{out} \cdot S_{out}^2 \cdot B] \cdot C_{in} \cdot k^2$ (number of output values $\times$ MACs per output value $\times$ 2)

- Linear $n_{in} \rightarrow n_{out}$ with bias:

$$FLOPs_{fc} = (2 \cdot n_{in} \cdot n_{out} + n_{out}) \cdot B$$

Layer	Substitution	=
conv1 7x7, 3→32, $S_{out} = S/2$	$2 \cdot 32 \cdot (S/2)^2 \cdot B \cdot 3 \cdot 7^2$	$2352 \cdot B \cdot S^2$
conv2 5x5, 32→64, $S_{out} = S/4$	$2 \cdot 64 \cdot (S/4)^2 \cdot B \cdot 32 \cdot 5^2$	$6400 \cdot B \cdot S^2$
conv3 3x3, 64→128, $S_{out} = S/8$	$2 \cdot 128 \cdot (S/8)^2 \cdot B \cdot 64 \cdot 3^2$	$2304 \cdot B \cdot S^2$
conv4 1x1, 128→256, $S_{out} = S/8$	$2 \cdot 256 \cdot (S/8)^2 \cdot B \cdot 128 \cdot 1^2$	$1024 \cdot B \cdot S^2$
conv5 3x3, 256→256, $S_{out} = S/16$	$2 \cdot 256 \cdot (S/16)^2 \cdot B \cdot 256 \cdot 3^2$	$4608 \cdot B \cdot S^2$
conv6 1x1, 256→512, $S_{out} = S/16$	$2 \cdot 512 \cdot (S/16)^2 \cdot B \cdot 256 \cdot 1^2$	$1024 \cdot B \cdot S^2$
fc1 512→256	$(2 \cdot 512 \cdot 256 + 256) \cdot B$	$262\,400 \cdot B$
fc2 256→100	$(2 \cdot 256 \cdot 100 + 100) \cdot B$	$51\,300 \cdot B$

Sum of all layers:

$$\begin{aligned}
 FLOPs(S, B) &= (2352 + 6400 + 2304 + 1024 + 4608 + 1024) \cdot B \cdot S^2 + (262\,400 + 51\,300) \cdot B \\
 &= 17\,712 \cdot B \cdot S^2 + 313\,700 \cdot B
 \end{aligned}$$

2. Memory (compared with `max_memory_allocated()`)

General formula – one activation tensor of C channels at side S_{out} :

$$\text{bytes} = 4 \cdot C \cdot S_{out}^2 \cdot B$$

Conv and BN each allocate such a tensor ($\times 2$ per block), MaxPool one, ReLU none (in-place).

Tensors	Substitution	=
conv1 out + bn1 out, $32 \times S/2$	$2 \cdot 4 \cdot 32 \cdot (S/2)^2 \cdot B$	$64 \cdot B \cdot S^2$
maxpool out, $32 \times S/4$	$4 \cdot 32 \cdot (S/4)^2 \cdot B$	$8 \cdot B \cdot S^2$
conv2 + bn2, $64 \times S/4$	$2 \cdot 4 \cdot 64 \cdot (S/4)^2 \cdot B$	$32 \cdot B \cdot S^2$
conv3 + bn3, $128 \times S/8$	$2 \cdot 4 \cdot 128 \cdot (S/8)^2 \cdot B$	$16 \cdot B \cdot S^2$
conv4 + bn4, $256 \times S/8$	$2 \cdot 4 \cdot 256 \cdot (S/8)^2 \cdot B$	$32 \cdot B \cdot S^2$
conv5 + bn5, $256 \times S/16$	$2 \cdot 4 \cdot 256 \cdot (S/16)^2 \cdot B$	$8 \cdot B \cdot S^2$
conv6 + bn6, $512 \times S/16$	$2 \cdot 4 \cdot 512 \cdot (S/16)^2 \cdot B$	$16 \cdot B \cdot S^2$
all conv-part activations		$176 \cdot B \cdot S^2$
head: GAP 512, fc1 256, fc2 100	$4 \cdot (512 + 256 + 100) \cdot B$	$3\,472 \cdot B$

$$\text{Memory}(S, B) = W + (12 + 176) \cdot B \cdot S^2 + 3\,472 \cdot B = 4.18 \text{ MB} + 188 \cdot B \cdot S^2 + 3\,472 \cdot B \text{ bytes}$$

3. Latency

Roofline per op plus a launch floor:

$$T_{\text{gpu}}(S, B) = \sum_{\text{ops}} \max(F_{\text{op}} / \pi, Q_{\text{op}} / \beta)$$

$$T(S, B, \Theta) = \max(t_{\text{launch}}, T_{\text{gpu}}(S, B)), \Theta = (t_{\text{launch}}, \pi, \beta)$$

- t_{launch} – CPU dispatch time of ~ 23 kernels.
- β – measured: device-to-device copy of 256 MB.
- π – fitted on end-to-end timings.

Regime: launch-bound if $t_{\text{launch}} \geq T_{\text{gpu}}$ else memory- or compute-bound by the larger term.

4. Energy (whole GPU)

$$E(S, B, \Theta_E) = P_{\text{static}} \cdot T(S, B) + P_{\text{dynamic}} \cdot T_{\text{gpu}}(S, B)$$